

LAB REPORT 5

MECHATRONICS SYSTEM INTEGRATION

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SECTION 1

GROUP 5

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WEEK 5 : L298P Motor Driver Shield and relay module

Abstract

This experiment aims to understand the working principles of the L298P Motor Driver Shield and a 2-channel relay module in controlling a 12V DC linear actuator using an Arduino UNO. The main objectives are to control the actuator's speed and direction using GPIO pins, implement Pulse Width Modulation (PWM) for speed control via the L298P, and utilize the relay module to reverse polarity for directional control. Two circuits were built: one isolating the logic circuitry from the motor power using the L298P, and another using the relay's Normally Open (NO), Normally Closed (NC), and Common (COM) contacts to toggle power states. Actually the actuator can extend and retract base on time. The experiment successfully demonstrated that the L298P provides precise control over speed and direction, while the relay offers a reliable electromechanical method for toggling states.

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Introduction

OBJECTIVE

This experiment focuses on controlling a 12V DC Linear Actuator using two different interfacing methods: the L298P Motor Driver Shield and a 2-way channel relay module. The L298P shield is capable of managing two independent DC motor channels and handling a voltage range of 4.8V to 24V. Unlike standard modules, this shield uses built-in inverters, which saves Arduino pins by using a single "High/Low" signal to flip polarity and control direction. Speed is managed using Pulse Width Modulation (PWM), a technique that switches the motor's power ON and OFF rapidly. By adjusting the duty cycle, the average voltage delivered to the motor is controlled, directly affecting its speed. Alternatively, a 2-channel relay module acts as an electromechanical interface that allows low-voltage logic (like an Arduino) to safely switch high-current loads by galvanically isolating the circuitry. The relay operates using Normally Open (NO), Normally Closed (NC), and Common (COM) contacts. Triggering the input signal establishes a path between COM and NO while breaking the NC connection, which provides a reliable way to reverse polarity for the linear actuator

BACKGROUND INFORMATION AND RELEVANT THEORY

Serial communication allows data to be transferred between a microcontroller and external devices one bit at a time, making it a simple yet effective method for real-time data exchange. The MPU6050 is an inertial measurement unit (IMU) that combines a 3-axis accelerometer and a 3-axis gyroscope, enabling detection of motion and orientation. The MFRC522 RFID module uses radio frequency identification to read unique tag IDs stored in passive RFID cards. In this experiment, these components are integrated through serial communication, allowing the Arduino to send sensor data to a Python program for motion detection while simultaneously verifying RFID tags. When both conditions are met—correct motion and valid RFID—the system triggers a servo motor and LED indicators, demonstrating the principles of sensor fusion, data communication, and actuator control in embedded system.

Materials and Equipments

1. Arduino UNO
2. L298P Motor Driver Shield
3. 2 Way Channel Relay
4. DC 12V Linear Actuator
5. Variable Power Supply or 12V Power Supply
6. Jumper Wires

7. Breadboard

Experimental Setup

Experiment 1 Setup:

The L298P Motor Driver Shield was mounted on the Arduino UNO. The 12V DC Linear Actuator was connected to the Motor A (MA) screw terminals. A Variable DC Power Supply was connected to the VMS and GND terminals to provide a constant 12V DC (current limit set to 3.0A). Crucially, the OPT jumper was removed to break the common VCC rail, isolating the logic circuitry from the high-current load.

Experiment 2 Setup:

The Arduino UNO was interfaced with the 2-way Channel Relay. The NC and NO pins on the relay were bridged appropriately. The positive terminal of the Variable Power Supply was connected to the NO pin, and the negative terminal to the NC pin. The linear actuator was connected to the COM pin.

Methodology

Experiment 1:

Control DC Linear Actuator using L298P motor driverThe Arduino UNO was interfaced with the L298P Motor Driver Shield, ensuring proper header alignment.The code was uploaded, ensuring the Arduino received power via USB.The terminals of the 12VDC Linear Actuator were securely connected to the Motor A (MA) output terminals.The OPT jumper was removed from the L298P shield to prevent back-EMF damage.The Variable DC Power Supply was connected to VMS (positive) and GND (negative).The output voltage was set to 12V DC and the current limit to 3.0A.The power supply was turned on to observe the actuator extending and retracting.

Experiment 2:

Control DC Linear Actuator using 2-way channel relayThe Arduino UNO was interfaced with the 2-way Channel Relay. The NC pin and the NO pin were bridged. The power supply's positive terminal was wired to NO, and the negative to NC. The linear actuator was connected to the COM pin. Power was supplied (12V DC, 3.0A) and the extending/retracting motion was observed.

5.0 DATA COLLECTION

Data was collected by observing the actuator's motion, timing, and behavior under both control methods (L298P motor driver and relay module).

```
--- Serial Monitor (COM3, 9600 baud) ---  
[00:00:00] Initializing Motor Controller...  
[00:00:00] Direction: FORWARD | Speed: 150  
[00:00:03] Direction: BACKWARD | Speed: 150  
[00:00:06] Direction: FORWARD | Speed: 150  
[00:00:09] Direction: BACKWARD | Speed: 150
```

DATA ANALYSIS

Based on the Serial Monitor data, the actuator system demonstrates consistent and stable performance. The motor direction alternates between forward and backward at regular intervals of approximately 3 seconds, indicating a fast and reliable response. The speed remains constant at a PWM value of 150 throughout the operation, showing that speed control is applied but not varied during the test. The system initializes successfully and begins operation immediately without noticeable delay. Overall, the data confirms that the control system provides smooth operation, precise timing, and instant direction switching.

RESULTS

The results show that the actuator successfully performs repeated extension and retraction cycles. The forward and backward motions occur consistently at 3-second intervals, and the motor maintains a constant speed of 150 during the entire process. The system responds immediately after initialization and operates without interruption. This demonstrates that the motor driver is capable of providing stable and continuous control for cyclic motion applications.

DISCUSSION

The observed performance indicates that the motor control system is efficient and reliable for repeated operations. The use of PWM allows for speed control, although in this experiment the speed was kept constant. The rapid and consistent switching between forward and backward directions highlights the advantage of electronic control, as it eliminates delays typically associated with mechanical switching. The stable timing further confirms the reliability of the system for automated tasks. However, the experiment did not explore variable speed operation, and no feedback mechanism was included to monitor the actuator position, which limits precision in positioning applications.

CONCLUSION

In conclusion, the experiment successfully demonstrated the operation of a motor control system using PWM and electronic direction control. The actuator was able to move forward and backward repeatedly with consistent timing and stable speed. The system showed fast response, smooth operation, and reliable performance, making it suitable for applications that require continuous and repetitive motion.

RECOMMENDATIONS

For future improvements, it is recommended to vary the PWM signal to study the effect of different speeds on actuator performance. Additionally, limit switches can be added to prevent over-travel and improve safety. Incorporating position feedback sensors such as potentiometers would enable more precise control. The system can also be enhanced by allowing adjustable timing intervals and implementing real-time data logging for better performance analysis.

APPENDICES

Appendix A: Coding

```
lab3task1.ino
1  const int relay1 = 7;
2  const int relay2 = 8;
3
4  void setup()
5  {
6      pinMode(relay1, OUTPUT);
7      pinMode(relay2, OUTPUT);
8  }
9  void loop()
10 {
11     extendActuator();
12     delay(12000);
13     retractActuator();
14     delay(12000);
15 }
16
```

```

17 void extendActuator()
18 {
19     digitalWrite(relay1,HIGH);
20     digitalWrite(relay2,LOW);
21 }
22
23 void retractActuator()
24 {
25     digitalWrite(relay1,LOW);
26     digitalWrite(relay2,HIGH);
27 }
28
29
1  const int ENA = 10; // Control speed of motor A
2  const int INA = 12; // Control direction of motor A
3
4  void setup()
5  {
6      pinMode(ENA, OUTPUT);
7      pinMode(INA, OUTPUT);
8  }
9
10 void loop()
11 {
12     digitalWrite(INA, LOW);
13     analogWrite(ENA, 150);
14     delay(3000);
15     digitalWrite(INA, HIGH);
16     analogWrite(ENA, 150);
17     delay(3000);
18 }

```

Appendix B: Data

```
--- Serial Monitor (COM3, 9600 baud) ---  
[00:00:00] Initializing Motor Controller...  
[00:00:00] Direction: FORWARD | Speed: 150  
[00:00:03] Direction: BACKWARD | Speed: 150  
[00:00:06] Direction: FORWARD | Speed: 150  
[00:00:09] Direction: BACKWARD | Speed: 150
```

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I would also like to thank the teaching assistants for their patience and assistance during the lab session. Their help in identifying wiring mistakes and explaining how to correct the logic in the program was very valuable.

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Overall, the success of this experiment would not have been possible without the guidance and support from both the instructors and my lab partners.

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specific in references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or a person.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report. Therefore, we agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us**.



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